

C7.4 Analysis

Analytical procedures

1. understand the difference between qualitative and quantitative methods of analysis;
2. understand that an analysis must be carried out on a sample that represents the bulk of the material under test;
3. understand that many analytical methods are based on samples in solution;
4. understand the need for standard procedures for the collection, storage and preparation of samples for analysis.

Chromatography

5. recall that in chromatography, substances are separated by movement of a mobile phase through a stationary phase;
6. know the meaning of the terms aqueous and non-aqueous as applied to solvents;
7. understand that for each component in a sample there is a dynamic equilibrium between the stationary and mobile phases;
8. understand that a separation by chromatography depends on the distribution of the compounds in the sample between the mobile and stationary phases;
9. understand the use of standard reference materials in chromatography;
10. describe and compare paper and thin-layer chromatography;
11. use the formula:
$$R_f = \frac{\text{distance travelled by solute}}{\text{distance travelled by solvent}}$$
12. understand the use of locating agents in paper or thin-layer chromatography;
13. recall in outline the procedure for separating a mixture by gas chromatography (gc);
14. understand the term retention time as applied to gc;
15. interpret print-outs from gc analyses.

Quantitative analysis by titration

16. understand the main stages of a quantitative analysis;
 - measuring out accurately a specific mass or volume of the sample,
 - working with replicate samples;
 - dissolving the samples quantitatively;
 - measuring a property of the solution quantitatively;
 - calculating a value from the measurements (IaS 1.2–1.4);
 - estimating the degree of uncertainty in the results (IaS 1.5–1.6);
17. recall that concentrations of solutions are measured in g/dm^3 ;
18. recall the procedure for making up a standard solution;
19. **calculate the concentration of a given volume of solution given the mass of solvent;**
20. **calculate the mass of solute in a given volume of solution with a specified concentration;**
21. recall the procedure for carrying out an acid-base titration using a pipette and burette;
22. substitute results in a given formula to interpret titration results quantitatively;
23. **use the balanced equation and relative formula-masses to interpret the results of a titration;**
24. use values from a series of titrations to assess the degree of uncertainty in a calculated value.